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Accurate evaluation of observability and redundancy in water production systems

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Abstract

Optimal sensor placement is vital for accurate and timely quality, safety, and research data. It is particularly important for water treatment facilities, where investment in infrastructure can be delayed by efficient use of installed sensors. Traditional placement based on immediate measurement needs can miss non-intuitive configurations that increase process-variable observability and sensor redundancy. We present a structural-labeling method based on symbolic evaluation of the linearized constraint Jacobian at an integer operating point: the Matrix Row Reduction for Observability and Redundancy (MRROR) algorithm. This algorithm simultaneously evaluates all constraints to assign observability and redundancy labels. We benchmarked MRROR against the GEneralized process Networks OBServability (GENOBS) and GEneralized process Networks REDundancy (GENRED) algorithms on two pilot water-treatment systems: a 6-stream membrane flowsheet with recycle and an 8-stream closed-circuit reverse osmosis flowsheet. Overall and dilute-solute component mass balances were used to establish bilinear constraints. By exhaustively enumerating sensor layouts and labeling observability/redundancy for each one, we identified Pareto-optimal designs under a multiobjective tradeoff between sensor cost, observability, and redundancy. GENOBS/GENRED misclassified observability in layouts requiring coupled hydraulic-component reasoning, while MRROR correctly labeled these cases and flagged variables with multiple finite solutions. Mislabeling affected 3% and 1% of the Pareto-optimal layouts for the 6-stream and 8-stream systems, respectively. The majority of MRROR-identified Pareto-optimal layouts were completely globally observable, meaning that only a single solution is mathematically possible for any operating point.

Keywords

Observability, Redundancy, Structural Labeling, Fault detection, Sensor Placement, Water Treatment

Introduction

Online sensors are central to the safe, reliable, and efficient operation of modern water production systems [1, 2]. Measurements of flow, pressure, water quality, and process performance support process monitoring, regulatory compliance, data reconciliation, fault detection, feedback control, and the development of digital twins [3–6]. As water and wastewater treatment facilities increasingly adopt automation and data-driven decision support, the value of a sensor network is determined not only by the number of installed instruments but also by whether those instruments provide enough information to estimate unmeasured variables and diagnose sensor or process faults [1, 4, 7]. Recent work on digital twins for wastewater treatment emphasizes the integration of real-time data, models, sensors, control methods, and Internet-of-Things infrastructure as core components of digital transformation in the water sector [5, 6]. However, adding more sensors does not necessarily produce more useful information [8]. Recent studies on measurement campaigns and data reconciliation in wastewater treatment have emphasized that sensor availability and cost are no longer the only limiting factors; rather, measurements must be selected so that the resulting data are informative, reliable, and compatible with mass-balance constraints [8, 9]. In this context, observability and redundancy are essential concepts. Observability determines whether unmeasured variables can be inferred from available measurements and process constraints, while redundancy determines whether measured variables remain observable if a sensor is removed or ignored [1, 2, 7]. These concepts are important because redundant measurements can improve data reconciliation, support fault detection, and increase confidence in soft-sensor estimates [3]. Sensor-placement methods based on structural observability and redundancy have therefore become useful tools for designing monitoring networks in water treatment and other networked process systems [10]. Prior work has formulated sensor placement as a multi-objective optimization problem that balances instrumentation cost, observable variables, and redundant measurements, often using Pareto-optimal layouts to identify practical trade-offs [1, 2, 11]. In wastewater treatment applications, such approaches have demonstrated that a limited number of sensors can be arranged to maximize information value rather than simply maximizing the amount of measured data [11, 12]. More recent work has continued to frame sensor planning and sensor network design around data reconciliation and mass-balance-based information recovery, reinforcing the need for rigorous methods that correctly classify what is and is not observable [8, 13]. Despite this progress, many structural-labeling workflows for process networks still rely on graph-based algorithms such as GENeralized process Networks OBServability (GENOBS) and GENeralized process Networks REDundancy (GENRED) [14, 15], and many more recent algorithms are based on GENOBS and GENRED specifically [1, 2]. These algorithms are attractive because they can classify observability and redundancy from flowsheet topology and sensor placement without requiring detailed numerical operating data [14, 15]. They have also been implemented in sensor-placement software packages, such as the Structural Observability and Redundancy (SOAR) toolbox, and used as the basis for Pareto-optimal sensor-network design [1, 2]. However, as demonstrated in this study, GENOBS/GENRED-type labeling can miss observable cases in bilinear systems when hydraulic and component balances must be solved simultaneously. This issue is particularly relevant for membrane-based water treatment systems that include recycle streams, splitters, mixers, and concentration constraints, as well as for data-reconciliation applications where bilinear mass balances can improve precision and gross-error detection [13, 16]. In such systems, some variables may be algebraically recoverable only through the coupled solution of flow and concentration balances, rather than through sequential evaluation of one balance type at a time. Consequently, missed observability cases can propagate into incorrect redundancy labels and altered Pareto front recommendations. Matrix-based methods provide an alternative route because they evaluate the solvability of the underlying equation system directly [17, 18]. Classical matrix-based reconciliation methods are well established, but previous implementations may be sensitive to numerical conditioning, floating-point errors, or assumptions about linearity [7, 18]. Therefore, there remains a need for a structural-labeling method that can evaluate coupled hydraulic and component constraints, avoid floating-point mislabeling, distinguish between unique and finite solution cases, and remain practical for exhaustive sensor-layout analysis in water production systems [19]. This paper introduces the Matrix Row Reduction for Observability and Redundancy method, MRROR, to address this need. The method evaluates the constraint equations by symbolically calculating the reduced row echelon form (RREF) of the unmeasured-variable Jacobian at a feasible integer operating point, enabling simultaneous consideration of hydraulic and component balances. MRROR was initially proposed as a method for evaluating pressurized water treatment systems, which incorporate nonlinear con-

straint equations that are not currently possible to evaluate using topological methods. During functional tests of MRROR against GENOBS/GENRED, an additional strength of MRROR was discovered. Unlike the GENOBS/GENRED algorithms, MRROR can identify variables that become observable only through coupled bilinear relationships. The method also supports redundancy labeling by testing whether a measured variable remains observable after its associated physical sensor is removed.

The objectives of this study are threefold: first, to demonstrate specific bilinear sensor-layout cases in which GENOBS/GENRED misclassify observable variables; second, to present MRROR as a matrix-based structural-labeling method capable of resolving these cases; and third, to quantify how corrected observability labels propagate into redundancy labels and Pareto-optimal sensor-placement results. These objectives are evaluated using two reverse-osmosis-inspired water production layouts: a minimal recycle membrane (MRM) system and a closed-circuit reverse osmosis (CCRO) system.

Methods

Definitions

Terms used throughout this paper are defined below, to eliminate ambiguity with conflicting definitions in other domains.

Flowsheet: A simplified diagram showing directional streams connecting process units such as membranes, pumps, and reactors.

Layout: A flowsheet, with the specific positions of hard sensors indicated on the stream whose variables are being measured.

Structural observability: A variable is *structurally observable* if it meets one of the following criteria:

- A sensor directly measures that variable.
- A finite set of solutions exists for the variable in the system of constraint equations, considering all measured variables.

Figure 1 illustrates this concept for a simple flowsheet containing two mixing nodes. In this example, the measured inlet and outlet flow rates around the first mixing node allow the intermediate stream flow rate to be estimated using a hydraulic balance. A second hydraulic balance then allows the downstream outlet flow rate to be inferred, even though it is not directly measured.

Structural redundancy: A sensor is *structurally redundant* if it meets both of the following criteria:

- It is a physical sensor, not a soft-sensor.
- The variable it is measuring remains structurally observable if the sensor is removed.

Structural redundancy is illustrated in Figure 2 using the same example flowsheet as Figure 1. In this case, flow-rate sensors are installed on the inlet and outlet streams of the first mixing node. If the outlet sensor is removed, the outlet flow rate can still be inferred from the two inlet measurements using the hydraulic balance. Therefore, the removed physical sensor is redundant because its measured variable remains observable without the sensor.

Surplus redundancy: Previously defined by Villez et al.[2]. It is not a label assigned to a sensor, but is instead used to categorize sensor layouts in which the number of observable variables and the number of non-redundant sensors is unchanged on the removal of at least one physical sensor.

Practical observability and redundancy: In this work, we focus on structural observability and redundancy. This means that quality of soft-sensor estimates is not considered. Doing so would lead to practical criteria for observability and redundancy, possibly including a quantified improvement of precision.

Global and Local Observability: For convenience, we define two categories for observable variables, inspired

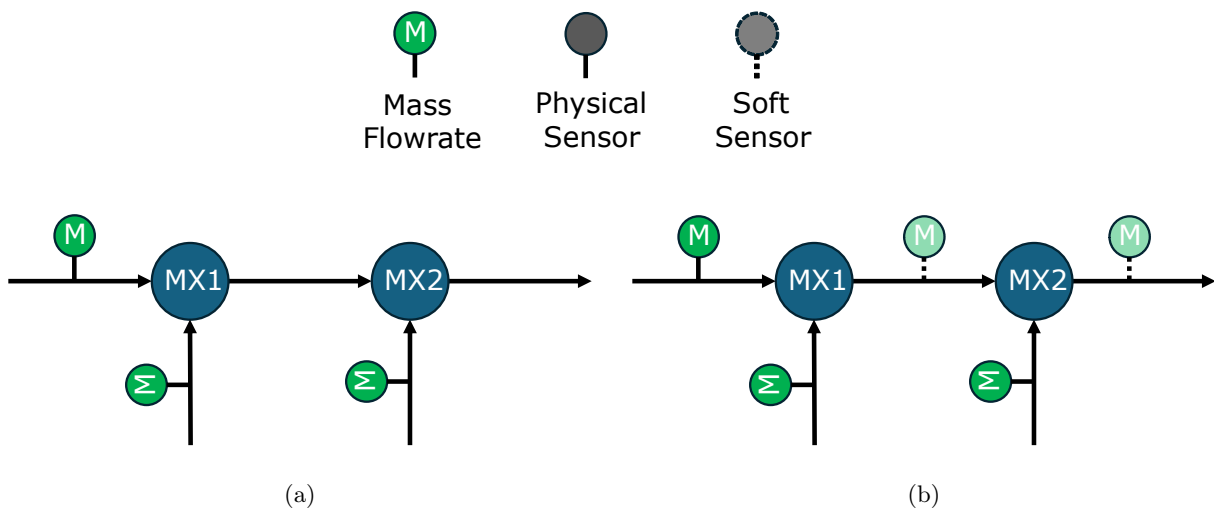


Figure 1: Example flowsheet showing (a) installed sensors and (b) soft sensors available through a hydraulic balance.

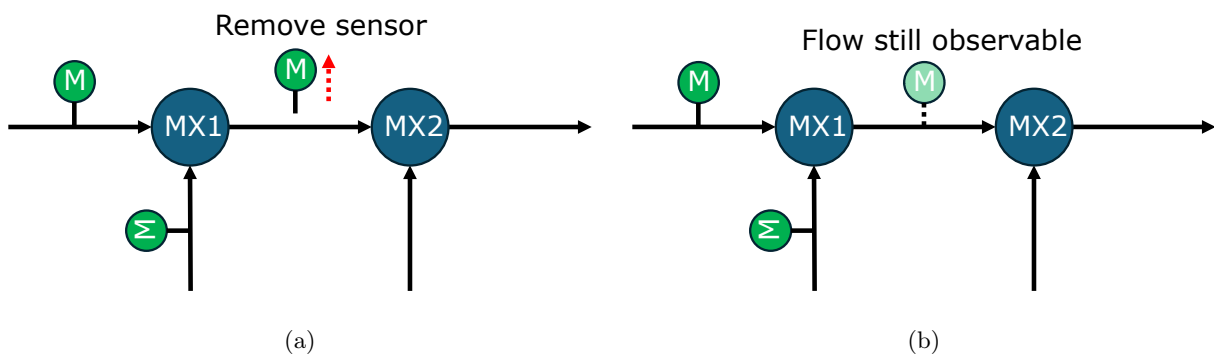


Figure 2: Example flowsheet illustrating redundancy with (a) removal of an installed sensor and (b) the soft sensor that remains after removal.

by writings from Walter et al.[20]. In some systems, an unmeasured variable may have multiple solutions. If the number of solutions is greater than one and less than infinity, it is locally observable. If it has only one solution, it is globally observable. If it has infinite possible solutions, it is unobservable.

Problem Statement

The GENOBS and GENRED methods were used to analyze process variable observability and sensor redundancy for every possible sensor layout on a given flowsheet, considering overall mass (hydraulic) balances and a component (species) balances as the only system constraints. The results of this analysis were compared with a new labeling method developed for nonlinear constraint systems, named the Matrix Row Reduction for Observability and Redundancy (MRROR) algorithm. An exhaustive search of all possible sensor layouts (see below) revealed that the GENOBS/GENRED algorithms are incorrect in the sense that observable variables as frequently labeled as unobservable. Pareto-optimal layouts were enumerated using both GENOBS/GENRED and MRROR, and the differences in the resulting Pareto fronts are discussed on detail.

Constraints

The variables of interest in this work are the mass/volume flow rate (m) and the composition of a dilute species (x). The assumptions and the particular hydraulic/component balances used for each process unit are included in the Supplementary Material. In general, the hydraulic balance over any node is given by Equation 1:

$$\sum_i^S a_i m_i = 0 \quad (1)$$

Where a_i is 1 if the stream flows into a node, -1 if it flows out, and 0 if it doesn't participate.

In this work, a bulk liquid and a single, dilute component will be considered. This simplifies notation and reduces the number of constraint equations. The general component balance equation is given in Equation 2

$$\sum_i^S a_i x_i m_i = 0 \quad (2)$$

Where x_i is the mass fraction of the dilute component in stream i . While it is generally uncommon that the mass flow rate and the mass fraction of a species will be sensed directly, for structural observability and redundancy, linear variable conversions (e.g. conductivity to salt concentration) are not necessary when defining the constraints. They are obviously important to reconciliation, however, so they should not be overlooked.

Flowsheets

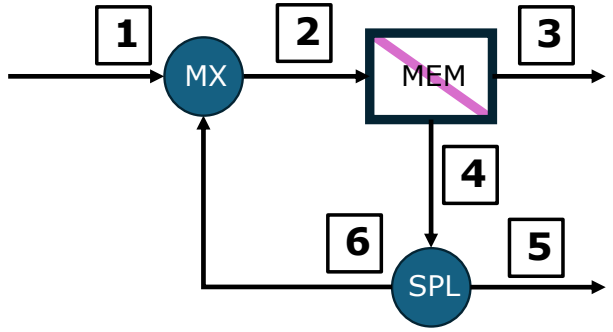
Two flowsheets are used to demonstrate the differences between the GENOBS/GENRED algorithms and MRROR. The first is an adapted process flow diagram for a pilot membrane filtration system, shown below. This flowsheet shall be referred to as the **Minimal Recycle Membrane flowsheet, or MRM**, throughout. This example was adapted from a pilot system maintained by the University of California Irvine (UCI) and the Orange County Water District (OCWD) [21]. **The physical system features a mixing node (MX), a Membrane unit (MEM) and a splitter (SPL).** As will be discussed later, the environment node (ENV) is used to handle flow into and out of the control volume. The MRM physical system, process flow diagram, and directed graph are shown below in Figure 3.

The MRM flowsheet was selected because it includes the following features:

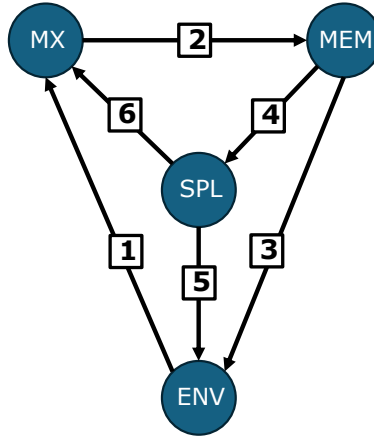
- A T-junction split (SPL in Figure 3b): Since no phase change, chemical process, or mechanical separation occurs, the inlet concentration and outlet concentrations from this node are equal.



(a)



(b)



(c)

Figure 3: Image of the MRM[21] physical system (a), process flow diagram (b), and directed graph representation (c).

- A recycle stream (stream 6 in Figure 3b): The concentrations exiting the retentate side of the membrane (stream 4) and the recycle stream have the same composition. The component balance over the MX node, therefore, can substitute the composition of stream 6 with the composition of stream 4.

It will be shown in the following sections how these features are exploited to highlight failure modes for the GENOBS algorithm, but in short, certain sensor layouts create a system where the compositions of streams 4, 5, and 6 are algebraically solvable by successive variable substitution in MRROR, but cannot be determined topologically using GENOBS.

The second process flow diagram analyzed is a closed-circuit reverse osmosis (CCRO) system, inspired by the experimental setup used in work by Chowdhury et al.[16]. A feed pump (PU1) and a recirculation pump (PU2) are present in this example, increasing the number of streams to 8, compared to the MRM flowsheet. The CCRO physical system, process flow diagram and directed graph are shown below in Figure 4. The motivation for including the CCRO flowsheet was its similarity to the MRM example, and the added

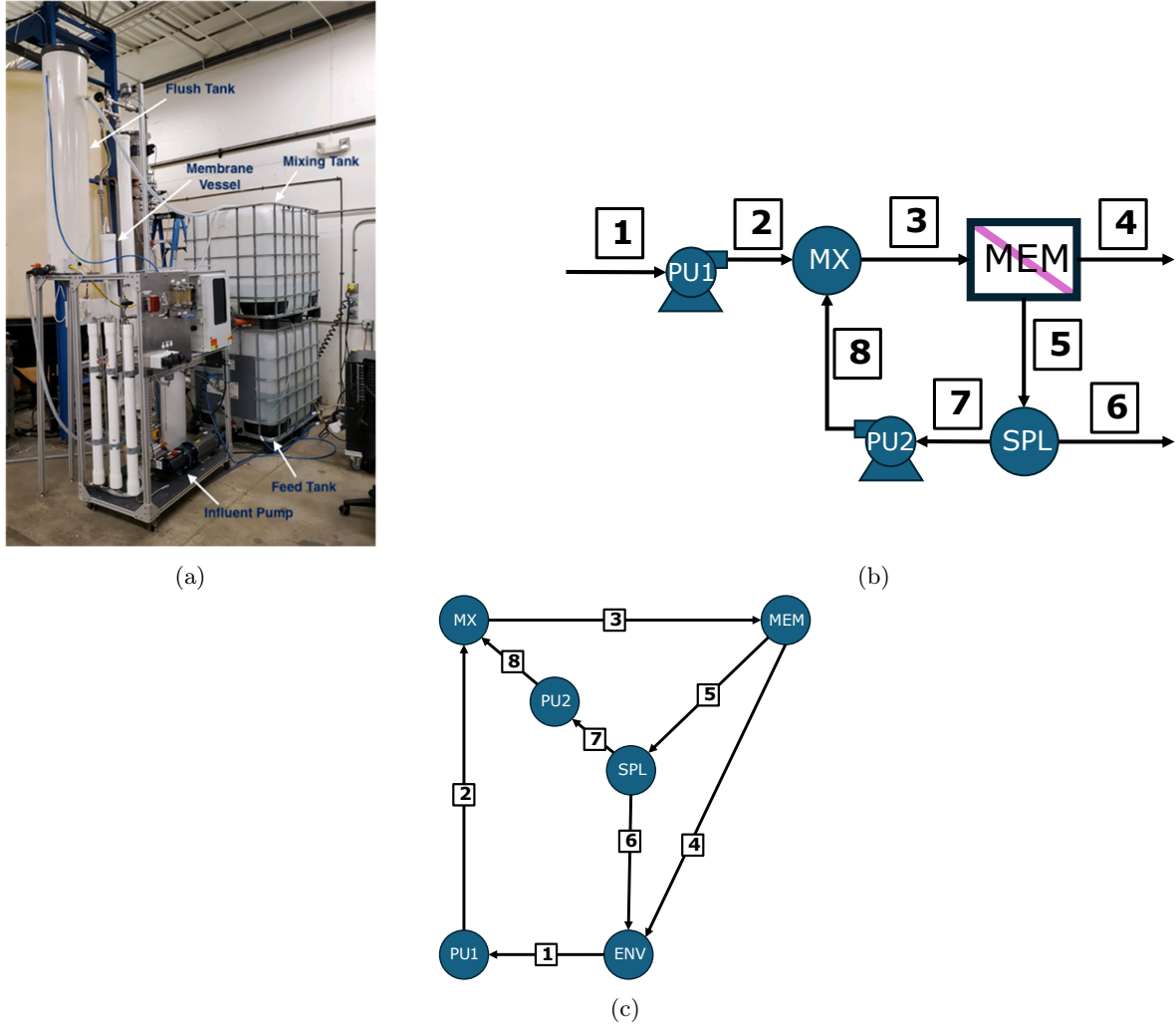


Figure 4: Image of the CCRO[16] physical system (a), process flow diagram (b), and directed graph representation (c).

complexity of 2 additional nodes. The additional two streams created by the inclusion of PU1 and PU2 means that 65,536 different sensor configurations exist for this flowsheet, compared to 4,096 for the MRM example. This makes for a fairer assessment of what happens to the Pareto front when the size of the system is increased.

Graph Theoretic Algorithms (GENOBS and GENRED)

Detailed treatments of graph-theoretic structural labeling are available in [7, 22]. The summary below focuses on the aspects needed to motivate the proposed method.

The graph in Figure 5 is an alternative representation of the flowsheet in Figures 1 and 2. The only difference is an additional environmental node (ENV), representing the world outside of the system boundary, has been added to convert the flowsheet into a mathematically valid graph. Graph-based labeling algorithms

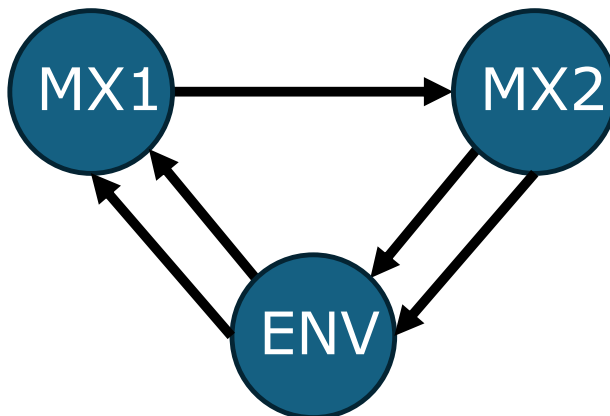


Figure 5: Directed graph for the example layout.

for linear (hydraulic) and bilinear (hydraulic and component) systems iteratively identify variables that can be estimated from other observable variables, using cycles in the graph. A cycle, in the context of a graph, is a path connecting one or more nodes, ending at the same node as the start, where each node on the path is only visited once [23]. For a simple mass balance, a mass flow rate is unobservable if there exists one or more cycles containing the flow rate that have only unmeasured flows [7]. This concept is illustrated using the prior example flowsheet in Figure 6.

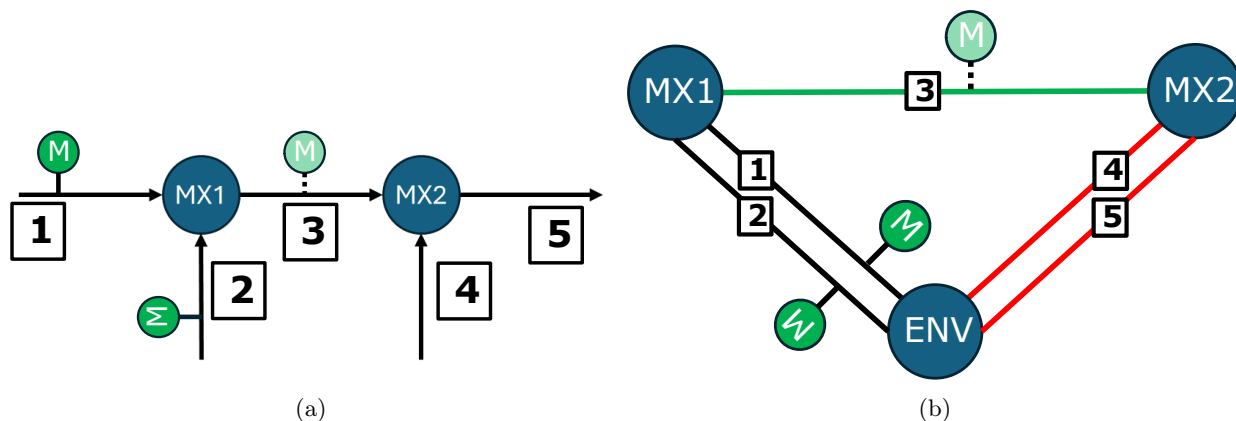


Figure 6: Example illustrating the flowsheet (a) and the graph (b) for the topological methods for structural labeling. The graph in (b) is colored to show that the mass flow rate for stream 3 is observable (green), because it does not lie on any cycles that contain only unmeasured flow rates. Streams 4 and 5 are unobservable (red) because they lie on a two stream cycle of unmeasured flow rates between nodes MX2 and ENV.

When concentration variables are included, the balance equations become bilinear because component-flow terms contain products of flow rate and concentration. GENOBS addresses this structure by alternating between hydraulic and component observability updates. This sequential propagation can identify variables that become observable after one subsystem has already been labeled. However, it does not perform algebraic elimination across the hydraulic and component balances as a single coupled system of equations. Conse-

quently, variables whose values can only be determined through simultaneous use of both balance types may remain unlabeled by GENOBS, even when the coupled equations admit a unique solution. This limitation is the primary motivation for the matrix-based method introduced in the following section.

MRROR Method

The proposed alternative method for structural labeling is referred to as Matrix Row Reduction for Observability and Redundancy (MRROR). It is based on established methods for observability and redundancy analysis based on linear algebra [24], and is inspired by prior work on reaction parameter estimation[19]. The defining trait of this method is the use of the reduced row echelon form (RREF) to determine observability. Let all steady-state process constraints be written as a vector-valued function,

$$\mathbf{g}(\mathbf{y}) = \mathbf{0}, \quad (3)$$

where $\mathbf{y}$ is the vector of all process variables. For nonlinear or bilinear constraints, MRROR uses the first-order expansion about a nominal operating point $\mathbf{y}_0$:

$$\mathbf{g}(\mathbf{y}) \approx \mathbf{g}(\mathbf{y}_0) + \mathbf{J}_0(\mathbf{y} - \mathbf{y}_0) = \mathbf{0}, \quad (4)$$

where $\mathbf{J}_0 = \partial \mathbf{g} / \partial \mathbf{y}|_{\mathbf{y}_0}$ is the Jacobian evaluated at $\mathbf{y}_0$. Partition $\mathbf{y}$ into measured variables $\mathbf{m}$ and unmeasured variables $\mathbf{u}$, with corresponding Jacobian blocks $\mathbf{J}_{\mathbf{m},0}$ and $\mathbf{J}_{\mathbf{u},0}$. The local linearized constraints can then be written as

$$\mathbf{g}(\mathbf{y}_0) + \mathbf{J}_{\mathbf{m},0}(\mathbf{m} - \mathbf{m}_0) + \mathbf{J}_{\mathbf{u},0}(\mathbf{u} - \mathbf{u}_0) = \mathbf{0}. \quad (5)$$

Collecting constant terms gives

$$\mathbf{J}_{\mathbf{m},0}\mathbf{m} + \mathbf{J}_{\mathbf{u},0}\mathbf{u} = -\mathbf{g}(\mathbf{y}_0) + \mathbf{J}_{\mathbf{m},0}\mathbf{m}_0 + \mathbf{J}_{\mathbf{u},0}\mathbf{u}_0, \quad (6)$$

which may be written compactly as

$$\mathbf{J}_{\mathbf{m},0}\mathbf{m} + \mathbf{J}_{\mathbf{u},0}\mathbf{u} = \mathbf{c}. \quad (7)$$

The observability decision is made by reducing the unmeasured-variable block $\mathbf{J}_{\mathbf{u},0}$, or equivalently the augmented matrix $[\mathbf{J}_{\mathbf{u},0} \ \mathbf{J}_{\mathbf{m},0} \ \mathbf{c}]$, to RREF. A representative RREF structure is shown in Figure 7[25].

$$\begin{array}{ccc} \begin{bmatrix} * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \end{bmatrix} & \xrightarrow{\text{Gauss-Jordan}} & \begin{bmatrix} 1 & * & 0 & 0 & * & 0 \\ 0 & 0 & 1 & 0 & * & 0 \\ 0 & 0 & 0 & 1 & * & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \\ (a) & & (b) \end{array}$$

Figure 7: Example of an arbitrary rectangular matrix (a) undergoing Gauss-Jordan Elimination to reduced row echelon form (b). Entries marked with a “*” can be either zero or nonzero.

A RREF is obtained via Gauss-Jordan elimination until the leading value in each row is one (known as a pivot), the values above and below each pivot are zero, and the pivot values increase in column index as the row index increases, creating the matrix shown in Figure 7a. Columns that contain a pivot are known as *basic*, and ones without are known as *non-basic* [25].

In RREF it is straightforward to evaluate rows that have a unique solution as ones that contain a pivot, and have zeros in every column except for the basic column. Following Gauss-Jordan elimination, a single “1” in a row with the remaining values being zero, is simply an equation with only one unknown in Equation 7, since $\mathbf{J}_{\mathbf{m},0}$, $\mathbf{m}$, $\mathbf{m}_0$, and $\mathbf{u}_0$ are all known values.

The matrix representation of $\mathbf{J}_{\mathbf{u}_0}$ in Equation 8 shows a situation where unmeasured variable u_4 can be uniquely solved:

$$RREF(\mathbf{J}_{\mathbf{u}_0}) = \begin{bmatrix} u_1 & u_2 & u_3 & u_4 & u_5 & u_6 \\ 1 & * & 0 & 0 & * & 0 \\ 0 & 0 & 1 & 0 & * & 0 \\ 0 & 0 & 0 & \underline{1} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (8)$$

Unmeasured variables that have a single, unique solution in $RREF(\mathbf{J}_{\mathbf{u}_0})$ are structurally observable. In the case of u_4 in Equation 8, the underlined 1 corresponds to a derived equation where u_4 is the only unmeasured variable present. Consequently, it is correctly labeled as structurally observable.

Sensor redundancy is determined by simply removing the column corresponding to the sensor from $\mathbf{J}_{\mathbf{m}_0}$, and placing it into the unmeasured Jacobian $\mathbf{J}_{\mathbf{u}_0}$, creating $\mathbf{J}'_{\mathbf{u}_0}$. This is visualized in equation Equation 9, using $\mathbf{m}_1$ to represent the column for the measurement to be checked for redundancy.

$$RREF(\mathbf{J}'_{\mathbf{u}_0}) = \begin{bmatrix} u_1 & u_2 & u_3 & u_4 & u_5 & u_6 & \mathbf{m}_1 \\ 1 & * & 0 & 0 & * & 0 & * \\ 0 & 0 & 1 & 0 & * & 0 & * \\ 0 & 0 & 0 & 1 & 0 & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & * \end{bmatrix} \quad (9)$$

The present implementation uses the SymPy RREF routine with exact arithmetic at integer-valued operating points. Exact arithmetic avoids tolerance-dependent rank decisions that can arise in floating-point matrix methods, while remaining sufficiently fast for exhaustive enumeration in the examples studied here.

Method Comparison

The Structural Observability and Redundancy (SOAR) toolbox[1, 2] utilizes the GENOBS and GENRED algorithms for labeling, with specific adaptations to enable efficient search for Pareto-optimal sensor layouts. The MRROR method was newly programmed in Python. All sensor placement permutations for all layouts tested were analyzed with both GENOBS/GENRED and MRROR. The code used for this study is available on request.

An exhaustive search of every possible sensor layout was performed for the MRM and CCRO flowsheets. Each stream could house a maximum of 2 sensors: One flow rate sensor, and one concentration sensor. For the six stream MRM flowsheet, the possible number of sensor layouts is 2^{12} , or 4,096 layouts. For the CCRO flowsheet, there are 2^{16} or 65,536 possible layouts. The results of this exhaustive search were then analyzed for Pareto optimality.

Pareto Front Analysis

Pareto-optimal layouts were determined using the same criteria as in prior work [1, 2]. The objective functions for number of sensors (f^c), observable variables (f^o), and redundant sensors (f^r) in Equation 10 through Equation 12 were used to determine Pareto-optimal layouts.

$$f^c = \sum_i^S \sum_j^V \mathbf{X}_{i,j} \quad (10)$$

$$f^o = \sum_i^S \sum_j^V (1 - \mathbf{Y}_{i,j}) \quad (11)$$

$$f^r = \sum_i^S \sum_j^V (1 - Z_{i,j}) \quad (12)$$

where $X_{i,j}$ is a binary vector indicating if a sensor is present (1) or not (0) for a given variable (V) and a given stream (S). $Y_{i,j}$ is a vector indicating observability of variable j in stream i, and $Z_{i,j}$ indicates the presence of a redundant sensor measuring variable j in stream i. For the systems analyzed in this work, the two variable types are overall mass flow rate (m), and solute mass fractions (x). Pareto-optimal[26] layouts were selected using the observability, redundancy, and cost objective function values for each layout. The objective function values for all layouts were iteratively compared, and answers which were outperformed by any single solution for all objective function values were excluded.

Solution Clustering Explanation

Sensor layouts were labeled with GENOBS/GENRED and MRROR, and categorized based on the following criteria:

- If all GENOBS and MRROR observability labels match, it is categorized in solution cluster 0.
- Additional clusters are grouped based on the variables that are labeled as observable through GENOBS. Each cluster corresponds to a set of layouts for which GENOBS produces the same observability labels.

Kretsovalis and Mah mathematically proved that their graph-decomposition algorithm determined observability unequivocally[14, 15, 22]. However, we demonstrate in this paper that GENOBS can fail to label observable variables as observable.

Our understanding of the GENOBS algorithm is that variables labeled observable through GENOBS are guaranteed to be globally observable. Our reasoning is as follows:

- The rules that label a variable observable based on nonlinear equations are limited to equations where only one variable has no label yet and all other variables are already labeled observable. Consequently, this necessarily leads to a unique estimate for the variable without label if unique estimates are available for observable variables.
- The rules that label multiple variables as observable at once are always based on linear equations, which are known to lead to unique estimates when the equation system is exactly determined (no redundancy) or over-determined (redundancy).
- Considering that only variables with unique estimates are labeled observable, it follows that GENOBS only labels globally observable variables as structurally observable.

Accordingly, all observable variables given layouts in cluster 0 (no difference between GENOBS and MRROR observability labels) are considered globally observable. For other clusters (1 and higher), we consider the variables that are labeled observable via GENOBS to be globally observable. The variables that are considered observable only via MRROR may be globally or locally observable. To distinguish global from local observability in these cases, we use symbolic analysis and express the MRROR-only observable variables as a function of the GENOBS-observable variables. If a single algebraic expression results for a variable, then that variable is labeled globally observable. If there are two or more solutions for a variable, then those variables are labeled as locally observable.

An example of this solution clustering, along with algorithmic details concerning variable labeling for global and local observability, can be found in the Supplementary Material.

Results and Discussion

The 4,096 different layouts for the MRM flowsheet and the 65,536 layouts for the CCRO flowsheet were analyzed for observability and redundancy labels using GENOBS/GENRED and MRROR. This comparison revealed a number of disagreements, which are characterized below. Exemplary cases are provided, starting

with the original sensor layouts for both flowsheets, where the GENOBS/GENRED and MRROR labels happen to agree. Pareto-optimal layouts using the results of the exhaustive search were determined based on the number of physical sensors, the number of observable variables, and the number of redundant sensors. Differences in the GENOBS/GENRED and MRROR Pareto fronts are also enumerated and discussed, with particular focus on the global/local observability of variables in the Pareto-optimal layouts.

Evaluation of Current Sensor Layouts

MRM Layout

The as-built layout for the MRM[21] flowsheet is shown in Figure 8. The GENOBS/GENRED and MRROR

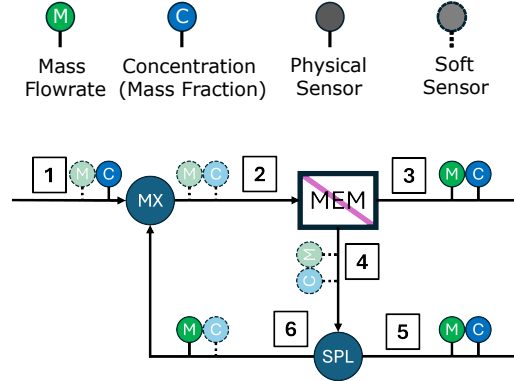


Figure 8: Existing sensor layout for the MRM system, showing physical and soft sensors. GENOBS/GENRED and MRROR lead to the same labels.

observability labels agree for this layout, and every unmeasured variable is observable. This can be shown by solving for each unmeasured variable sequentially:

$$\begin{aligned}
 m_1 &= m_3 + m_5 \\
 m_2 &= m_1 + m_6 \\
 m_4 &= m_2 - m_3 \\
 x_4 &= x_5 \\
 x_6 &= x_5 \\
 x_2 &= \frac{x_1 m_1 + x_6 m_6}{m_2}
 \end{aligned}$$

CCRO Layout

The as-built layout for the CCRO system, similar to the MRM system, is fully observable, as can be seen in Figure 9. The GENOBS/GENRED and MRROR labels are identical for this layout as well. Again, each

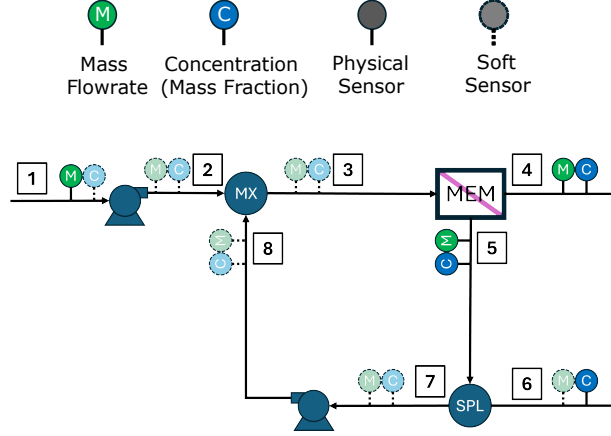


Figure 9: Existing sensor layout for CCRO system, showing physical and soft sensors. GENOBS/GENRED and MRROR algorithms lead to the same labels.

variable can be solved for sequentially, using the hydraulic and component balances:

$$\begin{aligned}
 m_6 &= m_1 - m_4 \\
 m_7 &= m_5 + m_6 \\
 m_8 &= m_7 \\
 m_2 &= m_1 \\
 m_3 &= m_2 + m_8 \\
 x_7 &= x_5 \\
 x_8 &= x_7 \\
 x_2 &= x_1 \\
 x_1 &= \frac{x_4 m_4 + x_6 m_6}{m_1} \\
 x_2 &= x_1 \\
 x_3 &= \frac{x_2 m_2 + x_8 m_8}{m_3}
 \end{aligned}$$

For both the MRM and the CCRO layouts, the unmeasured mass flow rates do not all need to be solved before solving for the concentrations. This was only done for clarity.

Exhaustive comparison

In total 4,096 layouts for the MRM flowsheet and 65,536 layouts for the CCRO flowsheet were labeled for observability and redundancy using GENOBS/GENRED and MRROR. The global observability and local observability were determined for each variable, and solution clusters were assigned based on which variables are labeled observable via GENOBS. Finally, layouts were evaluated to determine if they were Pareto-optimal. An overview of this analysis is shown in Figure 10 for the MRM flowsheet and Figure 11 for the CCRO flowsheet.

The labels for observability and redundancy on both flowsheets are identical for the majority of the possible layouts. The GENOBS/GENRED and MRROR labels are different for 46 layouts for the MRM flowsheet, 45 of which are correctly identified as Pareto-optimal by MRROR, but are considered suboptimal by GENOBS/GENRED. The CCRO flowsheet leads to 438 layouts with different labels, 407 of which were Pareto-optimal. There are no layouts leading to different labels for both the observability and redundancy.

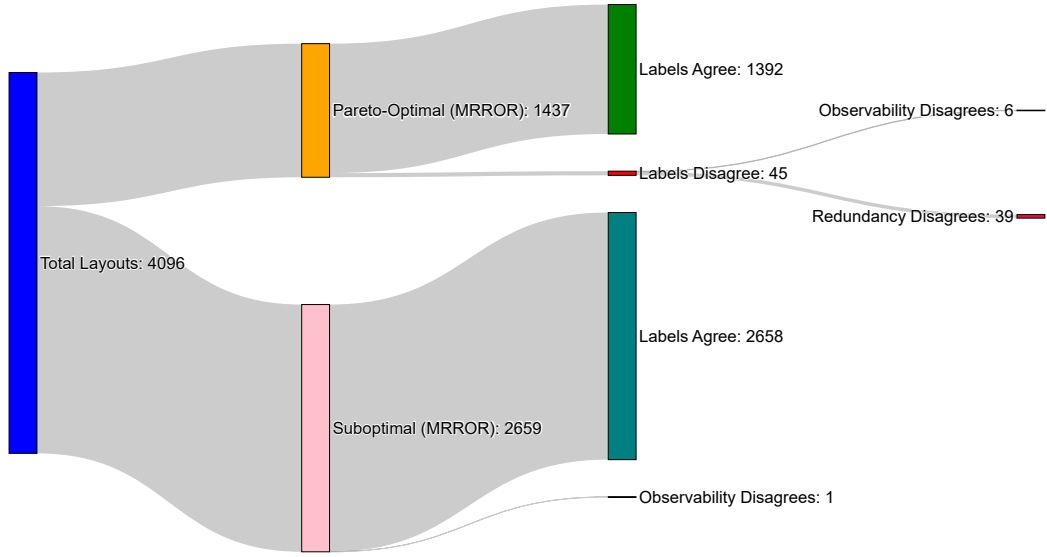


Figure 10: Sankey diagram showing the distribution of layouts for the exhaustive search of the MRM flowsheet. The total layouts were split into Pareto-optimal and suboptimal. These were further split into layouts where GENOBS/GENRED agreed with MRROR, and ones where the labels disagreed. Finally, where labels disagreed, they were categorized into whether the redundancy or observability labels disagreed.

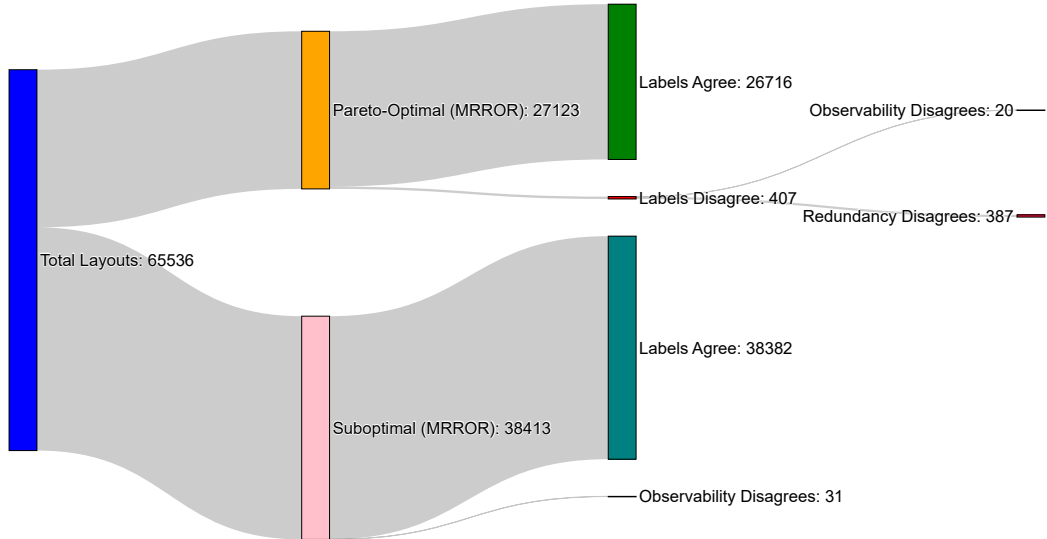


Figure 11: Sankey diagram showing the distribution of layouts for the exhaustive search of the CCRO flowsheet. The total layouts were split into Pareto-optimal and Pareto-suboptimal. These were further split into layouts where GENOBS/GENRED agreed with MRROR, and ones where the labels disagreed. Finally, where labels disagreed, they were categorized into whether the redundancy or observability labels disagreed.

Recall that a sensor is redundant if the variable it measures is still observable after the sensor is removed or ignored. Redundancy label disagreements between GENRED and MRROR are therefore equivalent to disagreements in observability labels for another layout where one or more sensors are removed. For this reason, we focus on differences in the observability labels.

Solution Clustering Analysis

Layouts where the redundancy labels did not agree were all grouped into solution cluster 0, as described in the Methods Section. All layouts where GENRED and MRROR disagreed on redundancy labels were fully observable via both GENOBS and MRROR. From a practical perspective, promoting a sensor from non-redundant to redundant is meaningful for error reduction and fault detection, but in either case the variable is still observable. This is also true for promoting a sensor from a single level of redundancy to higher orders of redundancy, i.e. cases where multiple soft sensors are discovered that can estimate an already measured variable. Therefore, the following cluster analysis will primarily be concerned with the 7 MRM layouts and 51 CCRO layouts where observability labels differed.

Solution clusters 1-4 contained the layouts where GENOBS and MRROR lead to distinct observability labels. Cluster 1 for both the MRM and CCRO flowsheets contained layouts where all observable variables are globally observable. Clusters 2-4 contain layouts where some observable variables are only locally observable. The solution clusters observed for the MRM layout are shown below in Table 1, along with the total number of layouts, number of layouts where all variables were globally observable, and the number of layouts that had some locally observable variables.

Cluster	Total layouts	All Variables Globally Observable	Some Variables Locally Observable	Pareto Optimal	GENOBS Observable Variables
1	4	4	0	3	$m_4, m_5, m_6, x_1, x_2, x_3$
2	1	0	1	1	m_3, m_6, x_1, x_2, x_3
3	1	0	1	1	m_2, m_5, x_1, x_2, x_3
4	1	0	1	1	m_1, m_4, x_1, x_2, x_3

Table 1: Cluster number, total layouts, number of layouts that are fully globally observable, number of layouts that have some locally observable variables, total Pareto-optimal layouts, and GENOBS observable variables for each each solution cluster in the MRM flowsheet.

The CCRO flowsheet also had 4 solution clusters when grouping the layouts where GENOBS and MRROR disagreed, as shown in Table 2.

Cluster	Total layouts	All Variables Globally Observable	Some Variables Locally Observable	Pareto Optimal	GENOBS Observable Variables
1	30	30	0	10	$m_5, m_6, m_7, m_8, x_1, x_2, x_3, x_4$
2	9	0	9	4	$m_4, m_7, m_8, x_1, x_2, x_3, x_4$
3	3	0	3	2	$m_3, m_6, x_1, x_2, x_3, x_4$
4	9	0	9	4	$m_1, m_2, m_5, x_1, x_2, x_3, x_4$

Table 2: Cluster number, total layouts, number of layouts that are fully globally observable, number of layouts that have some locally observable variables, and GENOBS observable variables for each each solution cluster in the CCRO flowsheet.

Worked examples are given below from solution clusters 1 and 4 of the MRM layout, to illustrate the algebraic solutions for a layout where all variables are globally observable, and a layout where some variables are only locally observable. Sensor positions for the remaining 4 MRM layouts and 51 CCRO layouts can be found in the supplementary material.

MRM Cluster 1: All Variables Globally Observable

The layout in Figure 12 had single solution for all unmeasured variables, making all variables globally observable:

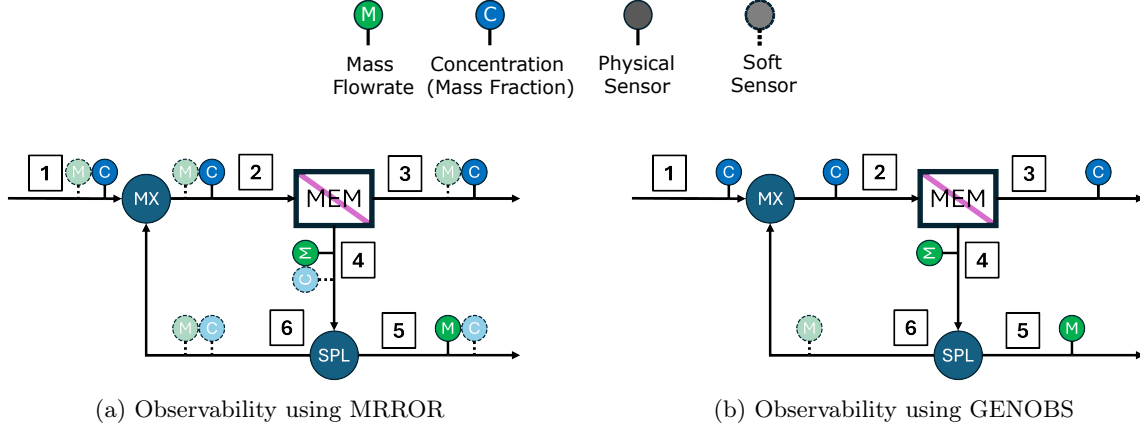


Figure 12: Sensor layout from MRM solution cluster 1 with the (a) MRROR reported observability and (b) GENOBS reported observability.

GENOBS is only able to identify m_6 as observable, but MRROR identifies all unmeasured variables as observable. Algebraically, the system can be solved for each variable, starting with x_4 :

$$x_4 = \frac{m_4 x_1 x_3 - m_4 x_2 x_3 + m_6 x_1 x_2 - m_6 x_1 x_3}{m_4 x_1 - m_4 x_2 + m_6 x_2 - m_6 x_3}$$

The remaining variables m_1 , m_2 , m_3 , x_5 and x_6 can then be solved for. We will simply use x_4 in place of the complete expression identified for brevity. An overall balance is used to estimate m_1 and m_3 next:

$$\begin{aligned} x_5 &= x_4 \\ x_6 &= x_4 \\ m_3 &= \frac{x_1 - x_4}{x_3 - x_1} m_5 \\ m_1 &= \frac{x_1 - x_4}{x_3 - x_1} m_5 + m_5 \\ m_2 &= m_6 + \frac{x_1 - x_4}{x_3 - x_1} m_5 + m_5 \end{aligned}$$

The equal concentration constraint between streams 4 and 6, combined with a balance over the MX and MEM nodes, make x_4 observable, which further makes the remaining mass flow rates observable. All variables in this layout are globally observable, since the expressions for each unmeasured variable will have a single solution at any given operating point.

MRM Cluster 4: Some Variables Locally Observable

The remaining 3 MRM solution clusters contain variables with multiple possible solutions given a single operating point. We will use the following layout from solution cluster 4 in Figure 13 as an example.

Unlike the previous layout, m_6 cannot be estimated using m_4 and m_5 . However, hydraulic and component balances can be constructed around MX and MEM, thus involving the flow rate and concentrations in streams 1, 3, 4, and 6. After substituting x_6 for x_4 , an algebraic solution can be found by solving for x_4 :

$$x_4 = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

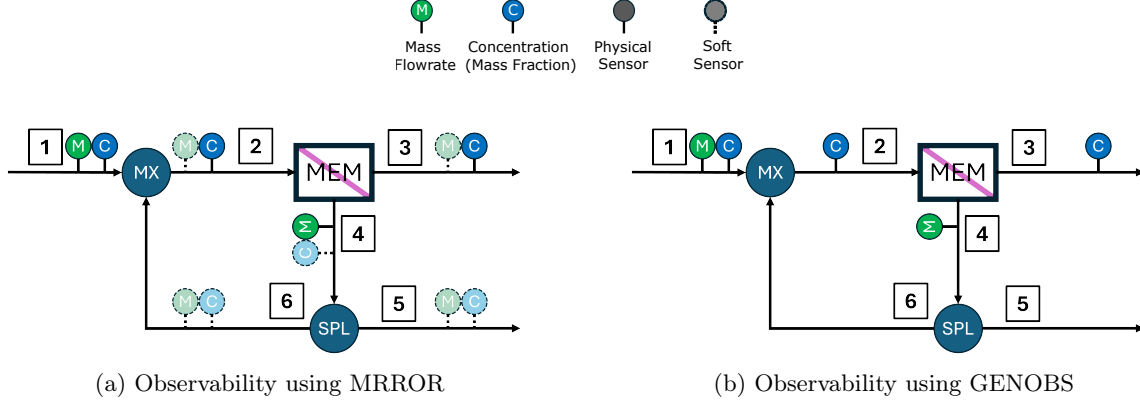


Figure 13: Sensor layout from MRM solution cluster 4 with the (a) MRROR observability labels and (b) GENOBS reported observability labels.

where the quadratic formula terms are as follows:

$$\begin{aligned}
 A &= \frac{m_4}{x_2 - x_3} \\
 B &= \frac{-m_1x_2 + m_1x_3 - m_4x_2 - m_4x_3}{x_2 - x_3} \\
 C &= \frac{m_1x_1x_2 - m_1x_1x_3 + m_4x_2x_3}{x_2 - x_3}
 \end{aligned}$$

Once x_4 is available, the remaining variables m_2 , m_3 , m_5 , m_6 , x_5 , and x_6 can be solved for:

$$\begin{aligned}
 x_5 &= x_4 \\
 x_6 &= x_4 \\
 m_6 &= \frac{x_1 - x_2}{x_2 - x_4} m_1 \\
 m_2 &= m_1 \left(\frac{x_1 - x_2}{x_2 - x_4} + 1 \right) \\
 m_5 &= m_4 - \frac{x_1 - x_2}{x_2 - x_4} m_1 \\
 m_3 &= m_1 \left(\frac{x_1 - x_2}{x_2 - x_4} + 1 \right) - m_4
 \end{aligned}$$

Every unmeasured variable in this sensor layout has a square root expression in its estimate, meaning either a positive or negative root could apply. This means that the practitioner may have to differentiate between 2 mathematically equivalent solutions for all unmeasured variables when performing reconciliation. In this layout, GENOBS did not label any of the unmeasured variables observable.

Pareto Front Analysis

The results from the exhaustive searches comparing GENOBS/GENRED and MRROR were used to determine the Pareto-optimal layouts for the MRM and CCRO flowsheets. The Pareto front for MRM flowsheet is shown in Figure 14b, with each Pareto-optimal point colored according to the number of observable variables and redundant sensors. The points on the Pareto front that were enlarged thanks to MRROR are highlighted, and the magnitude of MRROR's contributions to these points is shown in Figure 14a. In total, the newly identified Pareto-optimal layouts discovered using the MRROR represent 3% of the layouts on the Pareto front.

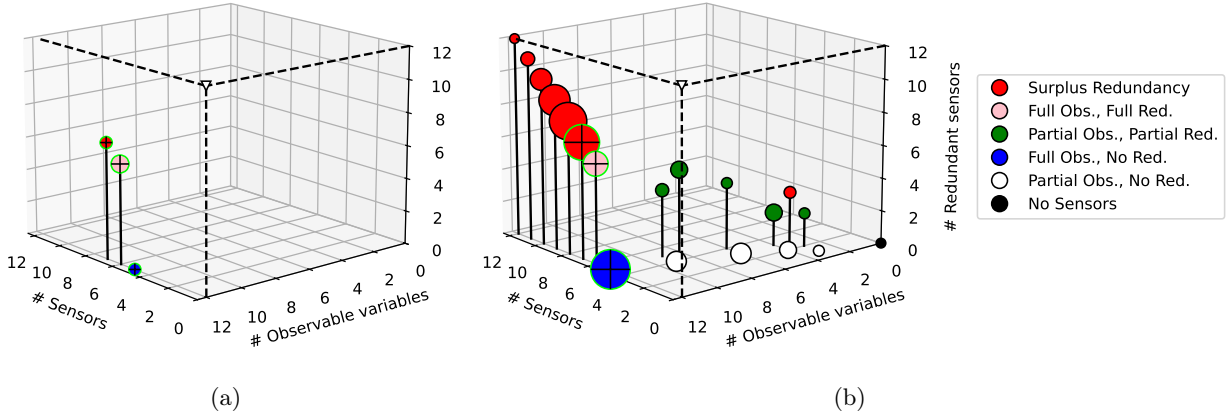


Figure 14: Minimal Recycle Membrane flowsheet – Plots showing (a) the additional layouts MRROR finds on the Pareto front and (b) the entire Pareto front for the MRROR method. Bubbles where more Pareto-optimal solutions were discovered have a green outline. The ideal solution (full observability, full redundancy, zero sensors) is indicated as a white triangle. The size of the bubbles reflects the number of layouts in each point on the Pareto front: (black) zero sensors; (white) incomplete observability, no redundancy; (green) incomplete observability, some or complete redundancy; (pink) complete observability, complete redundancy; (red) surplus redundancy. Points on the Pareto front that that contain solutions that are not identified via GENOBS/GENRED are highlighted lime green with a crosshair. The size of each point is relative to the number of Pareto-optimal layouts at that point.

An expanded version of Figure 14a is shown in Figure 15. The solution clusters that contribute to each bubble are shown in the bar chart on the right, along with the increase in redundant sensors or observable variables. Clusters 0 and 1 contain layouts where all variables are globally observable, while layouts 2, 3, and 4 all contain layouts that have some variables that are only locally observable. Figure 15 shows that most of the additions to the Pareto front are layouts where all sensors are redundant. Most of these layouts increase their redundant sensors by 1. Of the layouts with no redundancy, half are fully observable, and increase the observable variables by 6. The remainder increase observability by 7, but have some variables that are only locally observable.

The bottom left plot in Figure 15 shows that layouts for clusters 2-4 gain 7 observable variables using MRROR, compared to only 6 for cluster 1. All 6 layouts shown have 5 physical sensors in total (2 mass flow, 3 concentration). Pareto-optimal layouts in cluster 1 had one unmeasured variable that is considered observable via GENOBS, either m_4 , m_5 , or m_6 , while the remaining clusters did not.

407 new, Pareto-optimal layouts were discovered using MRROR on the CCRO flowsheet, which is approximately 1% of all the Pareto-optimal layouts. As was the case with the MRM layout, all of the additional Pareto-optimal layouts identified using MRROR are fully observable. The differing Pareto-optimal layouts for the MRM and CCRO flowsheets fall on the transition points from no redundancy to full redundancy, and full redundancy to surplus redundancy.

Most of the Pareto-optimal layouts identified with MRROR fall near the transition on the Pareto front between full variable observability and full sensor redundancy. This suggests that applying MRROR to existing sensor layouts that are close to (or already) fully observable may uncover fully observable or fully redundant configurations in flowsheets that were previously dismissed as only partially observable or as having no redundancy.

Figure 17 exhibits trends similar to the MRM flowsheet. Most of the MRROR additions to the Pareto front show an increase in redundancy, and most layouts increase their redundant sensors by 1. Like the MRM flowsheet, half of the layouts with no redundant sensors contain exclusively globally observable variables. Layouts with 9 additional observable variables all fall under clusters 2 and 4. An additional 10 observable variables are seen for 2 layouts that fall under cluster 3. Both of the layouts in this cluster have 5 physical

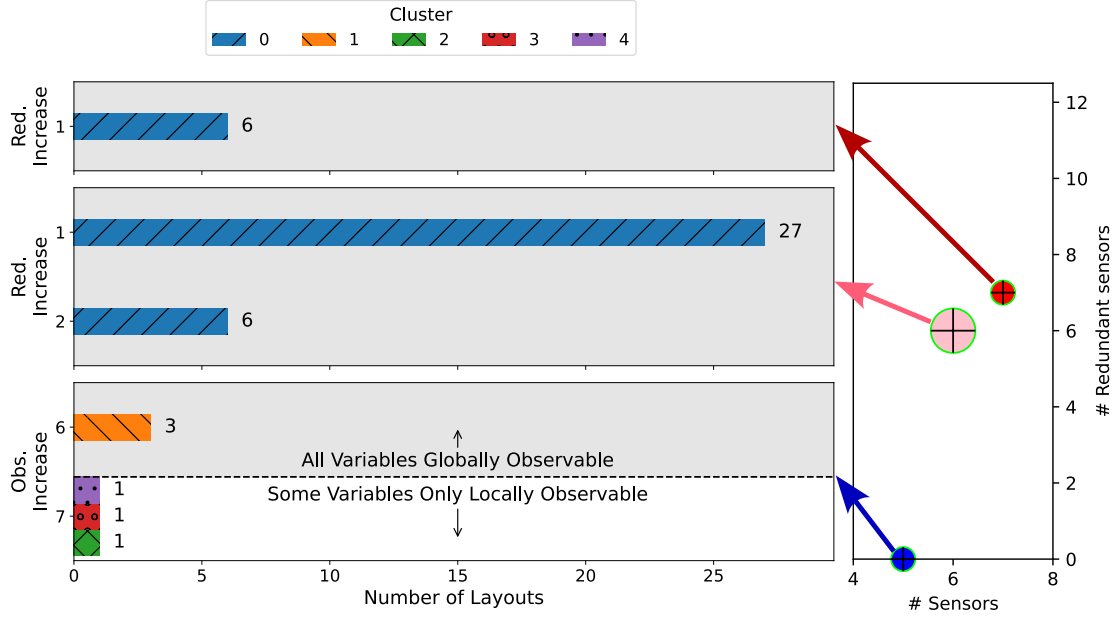


Figure 15: Increases in observability and redundancy observed for points on the Pareto-front for the MRM flowsheet. On the left, the increase in the number of redundant sensors or observable variables is indicated on the y-axis. The number of layouts that exhibited increased redundancy or observability is indicated on the x-axis. The color and pattern of the bar indicates the solution cluster that all of the contained layouts fell under. The right plot shows an expanded view of the Pareto plot in Figure 14, only displaying the points that gained layouts when MRROR was used.

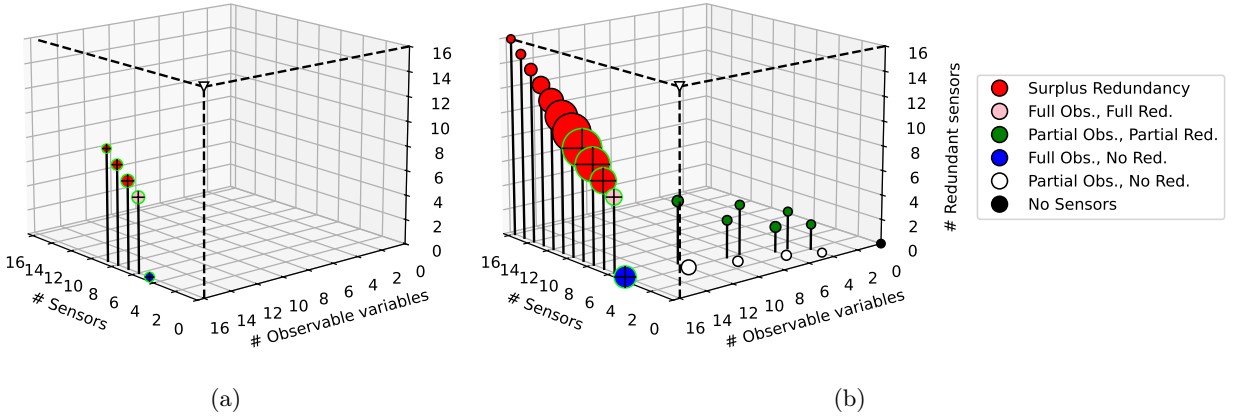


Figure 16: CCRO – Plots showing (a) the Additional layouts MRROR finds on the Pareto front and (b) the entire Pareto front for the MRROR method on the CCRO layout. The ideal solution (full observability, full redundancy, zero sensors) is indicated as a white triangle. The size of the bubbles reflects the number of layouts in each point on the Pareto front: (black) zero sensors; (white) incomplete observability, no redundancy; (green) incomplete observability, some or complete redundancy; (pink) complete observability, complete redundancy; (red) surplus redundancy. Points on the Pareto front that that contain solutions that are not identified via GENOBS/GENRED are highlighted lime green with a crosshair. The size of each bubble is relative to the number of Pareto-optimal layouts at that point.

sensors and only one unmeasured observable variable that is identified via GENOBS: either the inlet or outlet concentration from pump PU1.

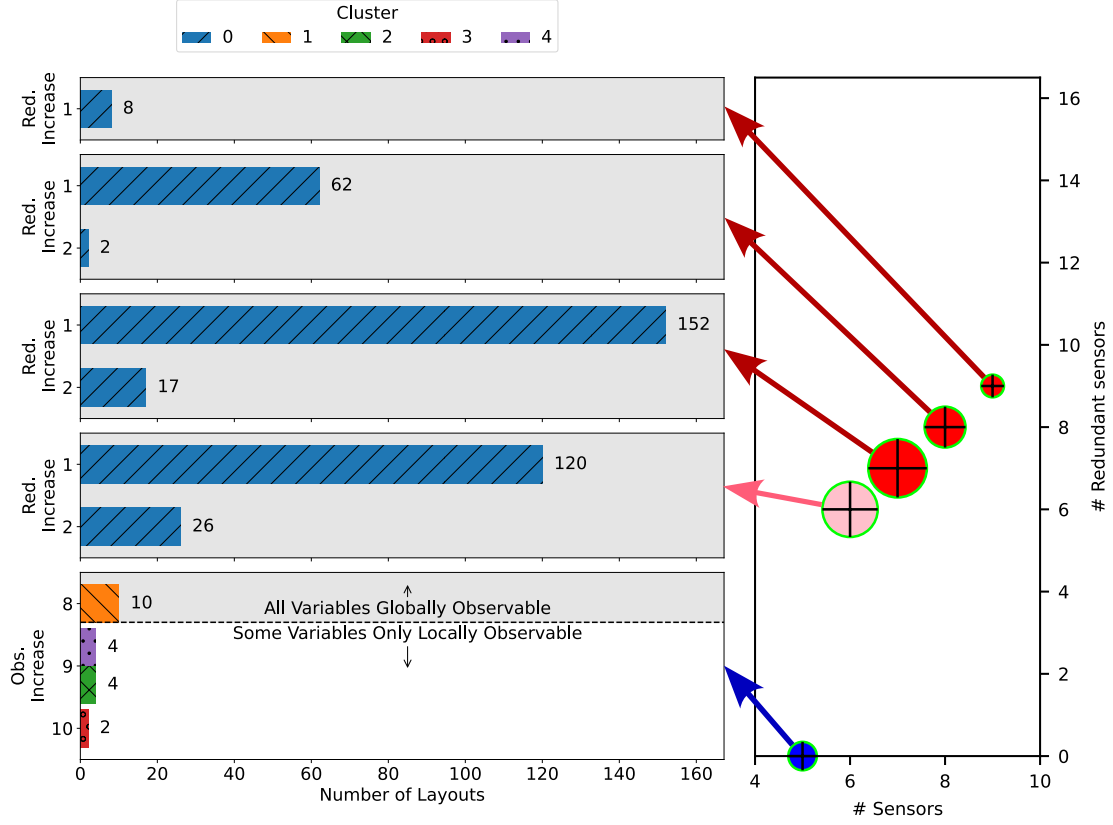


Figure 17: Increases in observability and redundancy for points on the CCRO flowsheet Pareto front. On the left, the increase in the number of redundant sensors or observable variables is indicated on the y-axis. The number of layouts that exhibited increased redundancy or observability is indicated on the x-axis. The color and pattern of the bar indicates the solution cluster that all of the contained layouts fell under. The right plot shows an expanded view of the Pareto plot in Figure 14, only displaying the points that gained layouts when MRROR was used.

Figures 15 and 17 reveal an encouraging result: most of the Pareto-optimal layouts discovered with MRROR contain layouts that are fully globally observable. This means that the estimates for the observable variables in these layouts are unique. Existing water treatment plant sensor layouts may have more observable variables and more redundant sensors than previously thought.

Method Comparison and Future Work

In the following sections, the results obtained via exhaustive use of GENOBS/GENRED and MRROR are discussed. A brief discussion of future direction for the MRROR algorithm will follow, and the challenges and opportunities that may be encountered.

Merit and drawbacks of the MRROR methods

Prior to this work, the primary motivation for developing the MRROR method for structural labeling was to analyze pressurized systems, which often include nonlinear constraint equations involving pressure (e.g. pump curves, membrane salt passage, valve friction losses, etc.). This paper demonstrates that the MRROR

algorithm is also useful for simpler bilinear constraint systems, where some layouts have observable variables that are mislabeled using the current formulations of GENOBS and GENRED.

One particular drawback persists, despite the advantages described. In its current formulation, it indiscriminately labels both globally and locally observable variables as simply observable. Our analysis is based on an automated computation of the symbolic solutions for each unmeasured variable in a given layout using SymPy. This approach is slow, and may be unmanageable for larger systems, so a more robust algorithmic approach is required for differentiating between global and local observability. This is not an insurmountable obstacle, as algorithms already exist that could in the future be used to more efficiently simplify and evaluate the system of constraint equations[27].

Merit and drawbacks of GENOBS/GENRED

An obvious drawback of GENOBS is that it mislabels some variables as unobservable. However, GENOBS does not mistakenly label any unobservable variables as observable. It is possible that the original intention of the GENOBS algorithm proposed by Kretsovalis and Mah[22] was to only label process values as observable if they have a single solution based directly off nodal balances.

GENOBS also avoid layouts with locally observable variables, like the ones demonstrated in the example case for MRM solution cluster 4. Depending on the operating point selected, this may not always be an issue. In cases where the distinction is nontrivial, it is up to interpretation if variables with multiple solutions qualify as observable at all. For our purposes, it is important that both local and global observability for all variables should be supplied to the user. They can decide if it fits their requirements for observability, given their equipment ranges, measurement units, conversions, etc. This necessitates using MRROR or similar methods, as topological structural labeling do not treat these cases the same.

Future Work

A possible future direction for this work could include adding a numerical degree of observability, based on the complexity of the equations used in reconciliation for global structural observability, and/or the polynomial order of the solution as rough approximation for the number of possible answers for locally structural observability. This would allow the user to retain layouts that included multiple solutions or complex reconciliation equations, but would penalize the observability objective function for answers depending on the number of solutions and equation complexity. Additionally, the integer operating point requirement for MRROR should be investigated further, to see if practical flowsheet size limits exist that would inhibit using this method for larger systems. Finally, MRROR should be tested for systems with more complex nonlinearity, to include constraint equations for membrane salt passage, pump efficiency curves, chemical reaction kinetics, compressible fluid flow, gas/liquid diffusion, and other constraints relevant to water treatment.

Conclusions

The following discoveries were made as a result of this work:

- A novel structural labeling method, named the Matrix Row Reduction for Observability and Redundancy (MRROR) method is shown to excel in structural labeling for two water production flowsheets, and identify additional layouts that are fully observable and layouts that are fully redundant that the GENOBS and GENRED algorithms miss.
- The historical GENOBS algorithm fails to properly label variables as observable. Specifically, GENOBS is incapable of labeling variables that are reconciled by simultaneously solving the hydraulic and component balances for a given flowsheet.
- Some layouts that were relabeled using MRROR had unmeasured variables with multiple solutions. New terminology was proposed to categorize these variables as either globally observable (single solution) or locally observable (multiple solutions).
- The layouts that GENOBS mislabels have significant impact on the Pareto front.

- The majority of MRROR-identified, Pareto-optimal layouts contain only globally observable variables, meaning they have only one possible solution for each variable at a given operating point.

Supplementary Material

- Supplementary material: Example calculations for solution cluster layouts, operating points used for Minimal Recycle Membrane (MRM) and closed circuit reverse osmosis (CCRO) flowsheets.

CRedit authorship statement

Christopher Blais: Conceptualization, Methodology, Software, Writing- Original draft preparation. Subrata Mukherjee: Conceptualization, Writing- Original draft preparation, Writing- Reviewing and Editing. Poo-ria Ghorbani Bam: Writing- Original draft preparation, Writing- Reviewing and Editing. Sally Ghanem: Writing- Reviewing and Editing. Diego Rosso: Conceptualization, Writing- Reviewing and Editing. Kris Villez: Conceptualization, Methodology, Software, Writing- Original draft preparation, Writing- Reviewing and Editing.

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